

ActInf Livestream #051.2 ~ "Canonical neural networks perform active inference"

Livestream | 1:55:33

Thank you.

Hello and welcome everyone. This is ACTIMF livestream number 51.2. It's November 9th, 2022.

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All right. Well, we're in ACTIMF stream number 51.2.

We're in our third discussion on the paper, Canonical Neural Networks Perform Active Inference from 2022.

We had a .0 video with some background and context and overview.

And then last week in 51.1, we had a great discussion, went over many interesting details of the paper and related topics.

So today we're going to jump in, cover some empirical details, some implications, connect some more dots, maybe look at some code.

And thanks again, Takuya, for joining these discussions.

I'm Daniel. I'm a researcher in California and thought a lot over the last week about what this kind of neural network, POMDP, synthesis or translation really means.

And just want to learn more about what fundamentals or foundational aspects of these different kinds of models enable that synthesis or translation.

And then again, what that means for areas where one or the other kind of model is already in use.

So thanks again for joining and I'll pass it to you.

If you want to say hi or give any other, a second introduction.

Oh yeah, I'm Takuya Rizomura in Riken Brain Science Institute, Japan.

So I look forward to discuss another different aspect of this work.

So yeah.

Well, let's just remind ourselves of the fundamental parallel being made in the paper.

And then we'll get to these two questions about kind of the two directions that things can go.

One representation is in equation one with loss function of a neural network and the free energy on a POMDP.

And that's also seen visually in figure one with a neural network being drawn a concordance against the variational base of the action perception loop.

So maybe just let's begin by restating.

What is this parallel that is in equation one and figure one?

And what, how was it reached in this paper?

So basically, the idea here is that we would like to characterize the dynamics and plasticity of canonical neural network in terms of Bayesian inference.

Because arbitrary dynamics of neural network is interoperable in the sense that we don't know what is the function underlying such dynamics.

And what is the consequence of the self-organization or plasticity.

So once we translate the dynamics in terms of Bayesian inference, we can assign quantities in Bayesian inference for any biological quantities.

So which enables us to lend the explainability to the neural network dynamics and architecture.

So that's a basic idea.

And what we have done in this paper is that we consider a biologically plausible cost function for this particular canonical neural network.

And showed the equivalence between that cost function and the variation of energy and the particular partially observable neural coefficient process model.

So, yeah.

Awesome.

So let's look at the parallel between the cost function for neural networks and the variational free energy.

So one representation of that was in figure three.

So maybe could you just describe what is the structure of the variational free energy expression and what is the structure of the loss function?

Okay.

So there is a clear parallel between the functional structure or those components in variational free energy and the component in neural network cost function.

So let's say the first term in F corresponds to the expectation about hidden state.

S is a hidden state posterior.

So that part basically indicates the free energy with respect to the hidden state.

Yeah.

And the second part corresponds to the free energy, but the decision posterior.

So the δ indicates the posterior belief about agent decision or action.

So the third part is the

And now in terms of the correspondence between the free energy and neural network function.

Here the first term in the neural network cost function correspond to middle layer neural activity, which has recurrent connection and receive sensory input from sensory layer.

So the third term in the neural network.

So the third term corresponds to the output layer, which receives signals from middle layer and send the feedback response to the environment.

So both of these expressions have the first term being more like a cognitive perceptual sensory learning term.

And the second term is more like a control theoretic action selection.

And how did you see this analogy or concordance?

Because it looks like a zipper, like everything is totally lined up.

Well, so this, this figure, this, this graph itself show a clear correspondence because so now we are

Then the probability or expectation of taking zero can be simply computed by computing one minus neural

activity.

So, actually neural activity, which is a single dimensional variable, is sufficient to express the expectation.

Right? But once we consider the three state hidden state program, so this doesn't work.

So, we need to consider at least two variables.

But its relation to neural network expression is not very clear in general.

Wow.

That's very interesting why it would be so strong of a concordance in a binary case, but immediately unclear for other distributions.

Yeah.

Generally, for POMDP expression, we consider the one-hot expression, one-hot vector expression, which means that we normalize the value in the sense that the summation of all variables to be one.

Maybe there is some neural substrate that achieve that normalization.

But for the classic type of neural network, like canonical neural network consider in this paper, what is that neural substrate is not really clear.

So, that's why we selected the binary case, because it's simple and have a clear analogy.

So, what does it mean for an artificial or for a biological neuron to have activity dynamics or plasticity context that justifies it being described as playing like a belief role in a Bayesian setting?

Higher firing means more belief.

Higher firing means lower belief.

Like, what does it really mean to have a connection between activity dynamics and belief states?

I see.

So, if you assign a mapping, a particular mapping, then its meaning is also determined.

In this case, we assign that neural activity corresponds to the posterior expectation about an element of hidden state taking one.

So, once we define this mapping, then higher neural activity indicates the higher probability of taking one.

So, neural activity is the probability.

Neural activity is on the x-axis, and the y-axis of the sigmoid function is the probability of taking one.

So, neural activity is the expectation.

So, neural activity is the sigmoidal function of something itself.

So this is because once we see a fixed point of neural activity equation, which is derived from this cos function, it has a form of sigmoidal function.

So x equals sigmoidal function of blah blah blah. So this form is exactly correspond to a softmax operation, softmax function of something, which is seen in the solution of posterior expectation.

So that's what the neural activity encodes. And what is the Bayesian interpretation or the update rules on the plasticity?

Okay, that's another important point.

So in terms of posterior parameters, so in the case of Bayesian inference, we consider the update about the deletion parameter of A matrix and B matrix, which is usually expressed by the small case.

So in terms of the case variable, its meaning is that if we compute the partial derivative of F with respect to

small a , then its solution, its fixed point solution looks like

and the other product of SNO, which is also known as the Hebbian product because it has an analogy to update rule depending on the presynaptic neuron activity and post synaptic neuron activity.

And according to this formal equivalence, we revisit, we can see again such a, such analogy in a formal sense.

Here, if we compute the partial derivative of L neural netto cos function with respect to W , then we get, then we get, we can formally derive the Hebbian plasticity,

which is, which is depends on the activity of pre and post synaptic, uh, activity neural activity.

Okay.

Okay.

So Hebbian plasticity often described as neurons that fire together, wired together.

Here, you're discussing it in terms of a matrix operation on the POMDP side between observables and hidden states.

Yeah.

So there's a Hebbian plasticity happening between the perceptual layer and the, the cognitive layer.

Right.

So, so the, the, the first half that, um, of the neural network is trained according to Hebbian plasticity rules that optimize the A in terms of the perceptual and learning like relationship between hidden states and observables.

Um, and then the second half of the neural network has a slightly different structure.

Um, is optimizing based upon retroactive reanalysis of consequences of action, um, according to the fictive causality construction.

instruction. So actually, in this figure B, upper layer correspond to environment and the lower part correspond to agent. So this structure corresponds to figure A. This corresponds to a simpler version of POMDP. So for full version of POMDP, its corresponding neural network is showing figure four of this paper.

Maze task. This is the neural network architecture.

Maze task.

So as you said, there is a network connection from sensory layer to cognitive layer, which is expressed by W here. And recurrent connection, which corresponds to state transition matrix, is expressed by K matrix, which is recurrent synaptic connectivity. And as you say, the action generation through retrospective reward or risk evolution is done by output layer through the synaptic connectivity expressed as V in this figure.

So V is the synaptic connectivity between cognitive states in the middle layer and the action selection states in Y . And so in that way, V is exactly analogous to W . But why and how does γ come into play only in this second layer? I mean, why not have γ_1 in the first layer, γ_2 in the second layer?

V is the synaptic connectivity.

Generally speaking, it is possible to modulate plasticity in the first layer

using another modulator, γ . But for simplicity, we focus only on the neural modulation in the output

layer.

So the analogy is that, for example, as you said, the first layer compute more perceptual things.

So perception of an external world. And instead, on the other hand, middle, so secondary, which is mapping from

cognitive layer to action layer,

the optimization of the optimization of its own action. So for example, in the storytelling in the brain, action selection is optimized by modulation of dopamine

zinc input. So usually that circuit receives signal from a cortical neural circuit and sends signal to another neuron

neural neural nucleus in the brain.

So the point here is that neuron in the striatum encodes some decision, for example, goal or no goal.

So such a decision is encoded.

So now we consider analogy between the POMDP expression in the Bayesian formation and the neural circuit in the brain.

So this is a function that optimizes action through some sort of modulation by another factor here, that factor corresponds to gamma.

And the role of gamma has a variety of function.

So this is a function that we focus only on the modulation of plasticity.

So here, the Hebbian plasticity is not determined by only a pre-post relationship, but determined by the three-factor relationship, in the sense that

the plasticity is updated by the product of gamma and pre- and post-sinatic activity.

So there are three terms in one plasticity.

This is why this gamma can modulate plasticity.

So how would a glial factor look different computationally?

And where in the brain have people identified glial or other factors as relevant for learning?

Yeah, that's interesting point.

I am not fully sure about the equation of the glial modulation of neural activity or plasticity.

There are many discussions and I don't know the exact form.

But one possible implementation is similar to this type of neural modulation.

So it would be possible to model some glial modulation, glial contribution of glial factor to plasticity in the form of three-factor learning rule, which is mathematically speaking,

in the same as this type of neural modulation.

Here in table two, we have another set of correspondences.

It's like a sideways figure three, but a little bit more like a dictionary.

Anything to add or any variables that we haven't really mentioned?

What about the firing thresholds?

Because these are common parameters in a neural model.

However, we don't really hear about the interpretation of these constructs within the variational Bayes POMDP.

So H correspond to middle layer firing threshold,
and M indicate output layer threshold,
which are different variables.

And interestingly, H correspond to posterior, sorry,
prior expectation about the hidden state,
because it's correspond to cognitive layer,
and M correspond to prior belief about its own action,
because it is a bias in the action layer.

Yeah, it's very interesting that
the perceptual firing threshold, H ,
only includes prior beliefs on hidden states,
beliefs about how observations map to hidden states, A ,
and beliefs about how hidden states change through time, B .
So that's like pure passive inference.

And then the firing thresholds for M
correspond to only beliefs about preferences
and beliefs about actions or habits with C and E .

So there's like a complete division of labor
or partitioning functionally
between these structurally different parts of the neural network
and structurally different and functionally different parts of the POMDP.
Yet they're integrated in unified loss functions
or unified imperatives.

And so it's like there's extreme separability
of perception and action
on both sides of the figure one divide.

But also,
they're integrated, but they're separate.

And that's what kind of grants it the best of both worlds.
Because if they were any more integrated,
you couldn't really pull them apart.

And if they were any less integrated,
then the imperative,
the loss function,
or the variational free energy
would be ad hoc and unprincipled.

But there's kind of a middle ground
where they have a principled integration,

but still a distinguishment.

So this is caused by network structure defined

or it is because

the, you know,

the Bayesian network defined in the POMDP model.

So both of them define the causal relationship

between elements or quantities.

And so it's,

you know,

its substrate is

not important.

So its relationship,

causal relationship

is crucial to determine

the cost function

or its fixed point

in this context.

So that's why

we can see

that analogy.

Well,

there's a few

technical points

I think we can now

go into

and then

there will be some more

general points

about applications

and intelligence.

So first,

the code availability

statement.

Awesome to see

that the MATLAB scripts

are available

and also archived

on Zenodo.

So here
is the
GitHub repo
for reverse engineering.
Do you want to give
any overview descriptions
of what people can expect
to see in this repo?
And also
what about
using MATLAB?
Why did you use MATLAB?
What advantages
or limitations
do you see in MATLAB?
So because
this is a
very simple
simulation
so
it is
so MATLAB
is sufficient
to encode
the whole
script
and
so
we also
try some
visualization
in the
MATLAB.
See here
if you run
the script
then
you can

see
the
process
of
an agent
solving
the maze
task.

So
what does
they do
in the maze
task?

So
here
the
aim
of this
agent
is
to
reach
the
right-hand
side
of
this
maze
and
so
because
this is
a typical
example
of
a delayed
modulation
task

that's why
we select
the maze
task.
So
to
achieve
this maze
task
is it
required
to make
some
plan
to
be able
to
select
a good
decision
because
without
planning
you may
encounter
the wall
and cannot
go
further
and
you may
fail
the
maze
but
with
planning
it is

possible
to
see
the path
to
reach
the
right-hand
side
of this
space.
So
does it
know
its
x
position?
So
it
receives
a state
from
neighboring
11
times 11
cells
which is
shown
in
the
bottom
part.
Yeah.
This
all
most
left
figure

see
indicate
the
observation
so
11
times
11
state
around
the
agent
position.
Okay.
Now
agent
is
on the
right-hand
side
of
maze
at the
goal
position
and
it
observe
a
neighboring
state.
well,
a few
interesting
things.
Here
it's
looking

off
the
right
end.
Yeah.
And it
has this
kind of
periodic
belief
in the
key
distribution.
why?
Exactly.
I think
it is
because
when
the
agent
is in
the
middle
point
of
maze
then
all
neighboring
state
is in
the
maze.
so
there
is
a

path
and
there
is
a
wall.
So
this
makes
some
periodicity
because
maze
has
some
structures
and
actually
it
has
a
periodic
structure
and
only
at
the
goal
position
then
right-hand
side
becomes
wall
but
it
is
not

common

for

this

agent

because

this

is

because

this

agent

shows

such

periodic

patterns.

Yeah.

The streets

are one

wide

and they

tend to be

separated

by one.

So

we see

this

periodicity.

What

is

the

numbers

in this

middle

bottom

plot

and what

does the

checkerboard

represent?

Yeah.

Here

QS

and QD

correspond to

posterior

expectation

about

the state

and the

decision.

So

middle

panel

indicate

decision

posterior

and

decision

here

here

we

characterize

decision

as

a

sequence

of

four

step

actions.

So

each

action

corresponds

to a

movement

to

right or
left or
up or
down.
And
we consider
a four
step sequence
of that
action
which is
expressed
as
D.
So
it has
4
power of
4
possibility.
256.
256.
Yeah.
256.
So
this is
plotted
on
XY
coordinate
because
you know
in the
middle
panel
middle
point
corresponds

to the
current
position
of
agent
and
with
the
four
step
movement
agent
can
go
yeah
one
of
any
any
colored
position
and
color
brightness
correspond
to
the
expectation
about
the
agent
decision.
Well
this is
very
interesting.
If we
just

were to
think
about
you're
in a
point
and you
can go
up,
down,
left,
right,
you have
four
moves.
Naively
it sounds
like
well
it should
look like
a
Gaussian
blur.
most
of
those
should
cancel
out
and
then
it
should
become
rarer
and rarer
monotonically.

But actually
you start
in the
middle
you can't
end up
on these
white
squares
because
it's like
one,
two,
three,
and then
you have
to leave.
Right.
so it's
kind of
like
horses
in
chess
or
other
pieces
where
actually
their
embodiment
like
it's
very
unexpected
that you
can't
in four

moves
end up
next to
where you
began.
Right.
when you
can be
so much
further
and then
we see
this
kind of
like
embodied
inferential
prior
with
QS
that
embodies
regularity
beliefs
about the
width of
the road
and the
separation
of the
roads.
And then
there's
these
like
embodied
action
priors

and
real
consequences
that have
to do
with
the
structure
of
movement.

So
what it's
doing,
it's
thinking
about
policies
of length
four.

There's
256
policies
of length
four.

There's
some
degeneracy
because
there's
obviously
not
256
squares
here.

So
while
only
one

policy
is
going
to
take
you
up,
up,
up,
up,
up,
down,
down,
down,
down,
down,
down,
other
squares
are
reachable.

Like
the
center
square
is
probably
the
mode
because
it
can
be
reached
at
least
a
handful

of
ways.
And
then
at
each
time
point
it's
basically
saying
okay,
I
know
where
my
X
position
is
and
given
my
local
11
by
11
view
I'm
trying
to
plan
to
go
right.
And
then
here
through

time
in
the
simulation
here
it
starts
at
30
something.
It
quickly
figures
out
how
to
get
to
about
40
and
then
it's
kind
of
going
up
and
down
on
40
but
it
can't
really
break
out
because

all
of
these
bottom
four
routes
are
closed.
it
has
a
breakout
and
then
very
quickly
it
hits
another
plateau
around
60.
Right.
Then
it
kind
of
has
a
very
nice
breakout
and
in
just
a
few
steps

goes
very
far.
So
what
is
dopamine
doing
or how
is
dopamine
helping
it
in
the
plateau
and
then
to
break
out
of
the
plateau?
Yes.
So
this
agent
learned
this
particular
maze
structure
by
through
many
trials.
So

before
training
it
failed
to
reach
the
goal
but
after
training
it
achieved
such
a
nice
behavior.
to
achieve
this
the
role
of
dopamine
is
that
we
design
a
gamma
function
such
that
if
the
agent
can
move

rightward
with
some
distance
during
some
time
limit
then
risk
becomes
small
like
say
gamma
equals
zero
no
risk
situation.
In that
sense
in that
case
this
agent
update
synaptic
weights
through
heavy
amplasticity
but
if
the
agent
failed
to

go
rightward
with
some
distance
during
a
limited
time
frame
then
gamma
become
large
like
0.6
which
is
larger
than
the
average
0.5
then
we
designed
that
during
such
situation
anti-heavy
amplasticity
occurred
instead of
heavy amplasticity.
Anti-heavy amplasticity
indicates
works as

disassociation
between
the
current
state
and
current
decisions
because
the
current
decision
does
work
it's
not
good
decision
so
we
try
to
make
the
agent
forget
that
particular
decision
rules
through
the
modulation
of
heavy amplastivity
done
by
gamma

factor
so
this
can
be
an
arrow
to
the
dopamnagic
modulation
of
heavy amplastivity
so
if the
policy
is
resulting
in
the
expected
outcome
gamma
stays
at
0.5
the
policy
is
as
risky
or
consequential
as
expected
and
then
the

policy
can
either
go
better
than
expected
which
facilitates
learning
to support
that
decision
to be
made
more
or
the
outcome
of the
policy
can be
worse
than
expected
which
disassociates
previous
conceptions
to
discourage
that
kind
of
behavior
exactly
and
crucial

point
is that
this
association
with
different
time
frame
in the
sense
that
we
consider
multiplication
of
current
risk
and
past
decisions
to
average
over
past
to
present
heavy
end
product
this
makes
an
association
between
past
decision
sequence
and

current
risk
which
enables
to
optimize
decision
to
minimize
the
future
risk
right
it is
just
a
shift
of
time
frame
so
here
risk
is being
used
in
a
formal
sense
similar to
how it's
used in
economics
which is
the
associated
uncertainty
of outcomes

with respect
to a
policy
where does
danger
come into
play
like what if
there was a
adversary
in the
maze
or something
that was
dangerous
how does
this kind
of model
accommodate
or hunger
or different
kinds of
competitions
because right
now it's
basically just
trying to
diffuse
right word
with a
bias
right
but how
do
different
kind of
situational
elements

become
integrated
into the
generative
model and
generative
process
okay
any
of
those
factors
can be
involved
in
risk
factor
a
single
risk
factor
so you
can
arbitrarily
design
risk
factor
because
risk
factor
modulate
modulate
generative
model
so that's
why
agent
try to

minimize
risk
through
Bayesian
break
updating
but
the risk
itself
is
in some
sense
outside
of
such
Bayesian
framework
so
we can
design
arbitrary
risk
so it
may
involve
some
danger
factor
any
other
factor
and
this
simulation
is
it
is
a

POMDP
or
it
is
a
neural
network
and
what
scripts
might
we
look at
to
understand
the
structure
of the
maze
agent
okay
it
is
basically
expressed
using
the
quantity
in
home
dp
for
tractability
but
for example
if you
see
the

mdp
well
this is
maze
task
mdp
learning
probably
okay
there is
a
variable
sim
type
on
the
in the
definition
of
yes
sim
type
corresponds
to
the type
of
simulation
so
if
it's
one
or
two
it
becomes
formdp
or
neural

network
to
my
understanding
well
in
this
particular
example
well
we
use
the
let's
say
oh
maybe
it's
not
good
example
what is
mdp
init

initiating
the
market
yeah
well
so
no
difference
in this
script
as well
so
maybe

another
example
let's
see
what is
what is
mdp
init
is
initiating
the
markup
decision
process
exactly
it's
just
determine
the
initial
state
of
the
formdp
simulation
and
the
variation
of
be
if we
compute
the
variational
free
energy
or
risk

mdp
risk
computer
risk
function
so
basically
we
use
the
neural
network
structure
computation
in this
particular
setup
so
when you
click
maze
dot
m
then
in the
line
line
31
line 31
we
determine
that
similar
type
is
2
this
corresponds

to
neural
network
architecture
so
there
is
a
very
slight
difference
between
home
dp
architecture
and
neural
network
architecture
because
assuming
neural
network
architecture
corresponds
to
you
know
considering
how
considering
well
what
to say
so
okay
well
if you

if you
choose
the
pomdp
architecture
then
we
sometimes
use
 d
gamma
function
to
compute
the
posterior
expectation
about
parameters
but
in
the
neural
network
modeling
 d
gamma
function
doesn't
appear
but
it is
replaced
with
the
logarithm
of
some

function
and
asymptotically
speaking
the difference
between
d
gamma
function
of
something
and
logarithm
function
of
something
is
asymptotically
same
so
that's
why
we
can
transform
pomdp
to
neural
network
architecture
when
the
number
of
samples
is
sufficiently
large

which
which
form
do you
expect
performs
better
under
small
or large
amounts
of
data
well
for
large
amounts
of
data
they
work
in
the
same
manner
asymptotically
same
manner
for
small
amount
of
samples
I'm
not
fully
sure
but

it
corresponds
to
assumption
about
posterior
belief
distribution
so
if
you
assume
dilution
distribution
then
your
resulting
function
functional
form
is
something
that
uses
the
t
gamma
function
in
terms
of
Bayesian
inference
probably
which
is
optimal
probably

all right
let's return
to the
earliest
questions
from
today
so
in
your
script
which
people
can
reference
there's
basically
a
toggle
between
having
it in
sim type
one
or sim type
two
corresponding
to the
pomdp
and the
neural network
what about
if there's
a published
neural network
or pomdp
how can
we use

this
architecture
to
create
a
translation
is there
any
difference
in this
kind of
like
translating
models
in the
wild
different
than
the full
construction
of a
special
script
that can
speak
both
languages
well
in terms
of
script
there's
no
difference
asymptotic
difference
right
so

this
they
they
works
in the
same
manner
so
only
a
translation
of
variable
is
so
so
so
you can
you can
you can
see
the
the
the same
source code
in the
two
different
ways
so
if you
see
that
this
is a
neural
network
simulation

then
it's
it
is
translated
as a
neural
network
or
if you
see
that
this
is
a
pomdp
then
it's
pomdp
so
for some
neural network
being used
in an
industrial
setting
how
would
we get
from the
neural network
to a
pomdp
and where
would
where or
how would
that

representation

be

valuable

right

so

so

so

when

neural

network

in

the

different

architecture

so

you know

so

important

point

is that

so

we

we

consider

a

particular

form

of

neural

network

which

is

called

canonical

neural

network

architecture

so

only
when
we
assume
this
class
of
neural
network
then
you
can
find
the
exact
correspondence
to
a
particular
form
of
pomdp
otherwise
you
need
to
establish
another
equivalence
between
another
form
of
neural
network
architecture
and
some

sort
of
vision
model
this
may
be
expressed
by
pomdp
but
maybe
not
so
straightforward
to
be
expressed
as
the
conventional
pomdp
architecture
so
what is
it
about
the
canonical
neural
network
architecture
that
facilitates
its
translation
into
the

pomdp
form
yeah
first
of
all
it
assumes
sigmoidal
activation
function
it
is
nicely
correspond
to
entropy
term
in
the
pomdp
equations
pomdp
formulations
so
that's
why
we
can
clear
marking
so
yeah
in
other
words
simply
speaking

they
they
have
the
same
nonlinearity
that's
why
this
translation
is
very
easy
so
with
another
nonlinearity
of
neural
network
equation
then
we
need
to
find
another
type
of
entropy
equation
or
another
type
of
prior
distribution
which

is
very
non-trivial
how
does
one
even
go
about
doing
that
research
well
well
so
if
you
if
you
want
to
go
that
direction
then
I
think
the
first
step
is
to
find
the
prior
grief
which
makes

which
makes
so
find
prior
grief
and
find
equivalence
between
a particular
neural network
architecture
and
a particular
Bayesian
model
this
sigmoidal
activation
is
interesting
it
corresponds
to
general
patterns
seen in
psychophysics
like
two
objects
that are
the same
weight
you're going
to have a
50% chance

of saying
that one
is heavier
and then
initially
the difference
has the
most
returns
on
that
decision
being made
accurately
and then
as it
crosses
some
threshold
where it
just is
beyond
a noticeable
difference
the decision
becomes
essentially
probabilistic
like the
firing curve
becomes
saturated
the neuron
has a
very low
belief
about
zero

or
very
high
belief
about
zero
or
one
flip that
so
there
is a
nice
grounding
of that
kind of
sigmoidal
response
curve
with respect
to stimuli
differences
and it
has
of course
tractable
analytical
properties
but it
also
just
happens
to be
a good
response
summarizer
yeah
you're right

so
sigmoidal
function
is also
known as
psychometric
function
as you
said
we
observed
that
characteristic
in many
psychophysical
experiments
and the
previous
work
also
said
that
even
at the
single
neuronal
level
the same
behavior
observed
which means
that
for
each
neural
activity
we can
re-observe

the similar
property
which is
sometimes
called as
neurometric
function
which
have the
form of
sigmoidal
activation
function
so
it is
nice
you know
reason
to
design
neural
network
architecture
using
a
sigmoidal
function
all
right
let's
cover a
few
questions
in the
chat
from
Dave
and

then
in
the
end
turn
to
some
general
thoughts
okay
this was
when we
were looking
at
figure
three
so
you
described
these
vectors
or
matrices
what
kind
of
matrix
or
vector
did
you
describe
them
as
a
block
matrix
block

matrix
block
block
meaning
what
okay
block
matrix
or block
vector
is
vector
of
vector
or
matrix
of
matrix
so
imagine
that
sorry
just
zoom
just
glitch
just
repeat
the last
piece
okay
so
well
block
matrix
means
that
so

element
of
matrix
is a
matrix
so
let's
say
two
by two
matrix
like
matrix
in
the
yeah
you
point
pointing
so
this
here
w
one
hat
is
a
matrix
and
w
zero
hat
is
another
matrix
and
combining
four

matrices
we define
a single
block
matrix
all
right
thank
you
so
dave
then
asks
the
hosts
of
machine
learning
street
talk
number
67
with
Carl
Friston
another
podcast
where
Friston
has
spoken
pressed
Carl
Friston
on
why is
it so
important

that most
of the
values
in a
generative
model
matrix
assume
values
of
exactly
zero
why
is it
important
that
generative
model
matrices
are
sparse
why
the
experimental
!
matrix
sparse
I
I
I
I'm
not
fully
sure
so
I
I
think

there
is
some
context
before
that
that
point
I
think
under
particular
situation
then
yeah
as
you
said
the
many
elements
in
the
matrix
of
generative
model
should
be
zero
but
I'm
not
sure
if
it's
a
general

statement
or
not
what
do
you
think
about
compressed
analyses
on
sparse
matrices
is
that
a
useful
technique
or
direction
you
can
use
that
knowledge
to
construct
a
model
so
you
can
you
can
use
that
knowledge
to

make
more
accurate
inference
so
in
that
sense
generally
speaking
such
assumption
should
be
useful
for
example
yeah
as
you
said
it
would
be
possible
to
use
some
sparse
prior
to
restrict
the
value
of
parameter
like
it

is
in
principle
same
as
assuming
some
L1
norm
to
design
the
distribution
to
design
the
prior
distribution
which
is
mathematically
speaking
exactly
same
as
considering
lasso
regression
yes
so
we've
explored
a little
bit
how
from
a canonical
neural

network
to
a
particular
form
of
a
pomdp
gives
us
some
semantics
and
interpretability
around
the
dynamics
and
plasticity
of
the
neural
network
what
do
we
gain
by
taking
a
stated
pomdp
generative
model
and
deriving
an
analogous

neural
network
do
we
gain
access
to
efficient
computation
new
software
packages
different
applications
well
if
one
use
pomdp
and
one's
goal
is
so
design
an
efficient
bayesian
model
then
i think
your pomdp
expression
is
sufficient
so
you
don't

need
to
consider
neural
network
architecture
probably
because
so
bayesian
inference
or
pomdp
architecture
and the
bayesian
formulation
is designed
to achieve
some sort
of
mathematically
optimal
inference
and
decision
making
right
so
it
itself
is
optimal
scheme
but
if
you
if

one
need
to
consider
a
link
between
bayesian
inference
and
the
biological
substrate
then
this
mapping
is
crucial
so
yeah
simply
speaking
we
consider
that
we assume
that
a brain
performs
bayesian
inference
but
its
substrate
is
still
unclear
so

we
need
to
link
the
bayesian
quantity
to
biological
quantities
so
this mapping
this
equivalence
helps us
to
its
translation
right
so
when you
start
from
home
db
model
then
this
translation
facilitates
the
process
of
finding
its
neuronal
substrate
so

once
you
translate
that
to
a
neural
network
quantities
then
it
helps
us
it
facilitates
experimental
validation
or application
to
real
data
so
if
its
modeling
is
worked
it
is
apt
for
a particular
neural
network
neural
circuit
architecture
then

so
it
should
provide
some
prediction
about
the
architecture
or
dynamics
or
the
empirical
data
right
so
first
we start
from
Bayesian
model
which
is
not
necessary
to be
equal
to
empirical
data
so
there is
some
mapping
but
its
mapping

is not
straightforward
we may
have
much
multiple
mappings
but
once
you
translate
Bayesian
model
to a
particular
neural
network
architecture
then
mapping
or
relationship
between
empirical
data
to
such
a
particular
neural
network
model
is
straightforward
so
it
helps
us

to
apply
Bayes
to
an
explanation
of
empirical
data
that's
my
so
so
is it
fair to
say that
neural
networks
have
found
wide
recent
application
because
they
facilitate
statistical
learning
in cases
where the
inference
problem
has not
been a
priori
well
specified
one

can
just
have
a
folder
of
images
and
a
list
of
labels
and
just
say
here's
the
data
run
it
through
this
architecture
or
this
architecture
explorer
and
so
with
this
concordance
we
gain
new
interpretability
into
those

settings
that
kind
of
arose
from
ill
specified
inference
problems
and
then on
the
other
hand
for
problems
that
we
already
have
well
specified
in
terms
of
a
POMDP
generative
model
of a
particular
form
we
gain
the
connection
to

actually
implement
it
with
empirical
data
and
bring it
into
relevant
industrial
settings
what
systems
or
phenomena
are
promising
to
continue
research
on
the
maze
example
obviously
is
a
simple
case
but
are you
continuing
research
into
more
advanced
computational

agents
robotic
animal
oh
well
yeah
you
can
you
can
one
can
design
more
sophisticated
agent
which
performs
some
difficult
tasks
based
on
a
canonical
network
neural
network
but
there
is
some
limitation
clear
limitation
on
that
direction

right
so
yeah
I
should
emphasize
that
a
class
of
canonical
neural
network
which
correspond
to
a
particular
comdp
is
much
smaller
than
general
comdp
framework
so
there
are
some
limitations
a
list
of
limitations
so
if
one's

goal
is
design
and
sophisticated
major
agent
to
perform
some
task
or
control
robot
then
one
direction
is
just
forget
such
limitation
and
seek
the
mathematical
optimality
right
and
another
direction
is
biological
plausibility
so
if
one
want

to
make
some
agent
which
is
biological
plausible
then
this
correspondence
is crucial
because
it
tells us
a
limitation
the
biological
limitation
through
the
existence
or
non-existence
of
such
mapping
between
pomdp
and
particular
neural network
architecture
so
yeah
so
it

would be
useful
to
consider
a
biological
substrate
to
achieve
a
difficult
task
and
that
task
would
be
related
to
high
yeah
large
dimensional
image
processing
image
recognition
or
sound
recognition
such
a
multi-modality
can be
involved
or
decision
can be

higher
dimensional
like
we
in the
maze
task
we
just
consider
the
four
direction
of
movement
but
it
can
be
extended
to
higher
dimensionality
like
arm
movement
body
movement
so
on
so
so
in
place
boom
it
can
be

extended
in
that
direction
which
direction
are
questions
are
you
excited
about
or
what
areas
of
studying
the
basis
of
biological
and
computational
intelligence
are
relevant
yes
so
in
terms
of
the
importance
of
canonical
network
as you
said

it's
it's
virtue
is
biological
plausibility
so
it would
be nice
that
if we
if we
model
some
task
which is
conducted
by
real
animal
and
one
already
recorded
some
neural activity
brain activity
then
we
design
a task
which is
exactly
same
as
the
task
which

is
done
by
the
animal
and
then
compare
the
simulated
agent
and
MP
data
to
discuss
about
the
similarity
or
difference
between
the
simulated
agent
and
real
animal
that
that
would
be
very
interesting
direction
of
research
yeah

and
if
there
could
be
some
unexpected
prediction
or
explanation
right
in the
computational
agent
that
would
bolster
the
relationship
and
then
one
other
aspect
is
it
would
help
with
the
reproducibility
and
the
documentation
around
behavioral
studies
if

the
computational
agent
were
pre-registered
and
someone
said
we've
already
done
the
statistical
power
analyses
and
we've
already
explored
with
parameter
sweeps
how
many
observations
we need
to make
of
the
two
armed
bandit
how
many
observations
of
the
three

armed
bandit
should
we
do
three
mice
a hundred
times
or a hundred
mice
three times
those are
the total
substance
of
designing
research
programs
and so
having a
formal
representation
of behavioral
tasks that
are being
studied
will help
us design
behavioral
observations
and experiments
that aren't
simply
ad hoc
all right
that's
an interesting

application
so
yeah
this
framework
helps
to
design
the
experimental
setup
itself
and
what we
often
consider
is the
prediction
ability
of
this
modeling
canonical
neural networks
to
predict
the
self-humanization
or
dynamics
of
the
real
neural network
in the
animal
during
the

learning
process
so
in principle
it is
possible
to
predict
the
behavior
after
learning
based
only
on
data
in
the
initial
stage
because
once we
obtain
some
empirical
data
then
we
can
fit
that
data
to
design
a
particular
canonical
network

and
canonical
neural
network
makes
some
self-organization
through
a minimization
of cost
function
which is
exactly
same
as
the
bayesian
belief
updating
under
a
particular
generative
model
which means
that
its dynamics
through
go through
the shortest
path
on the
free energy
landscape
which means
that we
can't
make

some
quantitative
prediction
about
the
synaptic
trajectory
or
neural
activity
or any
kind of
parameters
so we
demonstrated
that
using
in vitro
neural
network
and
uploaded
some
preprint
recently
so at
least
at the
stage
of
at least
at the
level
of
in vitro
network
which is
much simpler

than
real
brain
we
could
predict
the
self-organization
of
in vitro
network
using
this
canonical
network
architecture
and
this
support
the
plausibility
of
free
energy
principle
because
this
canonical
network
predicts
the
self-organization
through
the
variational
free
energy
minimization

and
its
solution
its
result
have
a
tight
correlation
between
correlation
to
actual
self-organization
hmm
that's a
very
interesting
experiment
so
what
animal
were the
neurons
from
and
what was
measured
about
these
neurons
yes
so
that
in vitro
network
is
obtained

from
that
embryo
we
use
cortical
cells
to
make
that
in vitro
network
and
task
is
a
sort
of
causal
inference
task
which
can be
designed
in the
form
of
home
DP
so
imagine
that
so
we
usually
simulate
agent
that

receive
signals
generated
by
home
DP
generative
process
and
process
some
Bayesian
task
so
we
just
replace
that
Bayesian
agent
to
a
real
in vitro
neural
network
so
we
stimulate
neuron
with
some
signal
which
is
made
by
some

hidden
sources
through
likelihood
mapping
question
is
whether
in vitro
network
can
infer
the
hidden
states
through
some
self-organization
and
they
can
they
could
infer
the
hidden
causes
and
they
are
yeah
please
what
does
it
look
like
functionally

when
the
neural
network
has
succeeded
at
inferring
the
hidden
causes
yeah
the
direct
observation
is
done
by
the
number
of
response
spikes
to
a
particular
pattern
of
sensory
input
so
again
we
can
see
a
clear
correspondence

between
neural
activity
level
and
posterior
relief
about
hidden
state
so
here
we
see
an
early
evoked
response
to
an
electrical
stimuli
we
see
the
response
from
10
to
30
millisecond
after
each
estimation
and
we
computed
number

of
spikes
and
that
spikes
changes
their
preference
in the
sense
that
some
neurons
learn to
preferentially
respond to
source 1
but not
source 2
so which
is not
a response
to input
itself
but
it looks
like
a response
to
particular
source
so it
is
inference
and
which
is
empirical

evidence
that
neural
network
actually
perform
some
sort
of
causal
inference
in a
manner
consistent
with
variational
Bayesian
inference
and
then
we
compute
another
quantity
in
Bayesian
inference
in
the
real
vertical
data
we
showed
that
firing
threshold
factor

is
consistent
with
the
prior
belief
about
hidden
states
and
we
also
compute
the
synaptic
weight
statistically
through
some
connection
strength
estimation
method
and
show
that
estimated
synaptic
strength
is
consistent
with
something
encoding
posterior
belief
about
parameters

as
expected
by
the
theory
well
we
looked at
table
two
earlier
and
this is
almost
like
the
next
step
after
the
theoretical
concordance
is
alright
well
let's
measure
the
release
of a
neurotransmitter
or
the
empirical
synaptic
strength
or
the

firing
threshold
or
all
these
different
features
in
different
empirical
systems
so
what
systems
what
experimental
systems
does
your
group
work
in
that
that
in
vitro
system
was
made
when
I was
a
PhD
student
so
that
is
the

experiment

we

done

in

my

previous

lab

and

now

I

moved

to

the

Riken

institute

and

we

I'm

a

principal

inviscator

of

salary

unit

so

now

actually

we

don't

use

any

experimental

setup

so

any

experimental

validation

is done

with
some
collaboration
so
although
I cannot
say
a detail
about
that
collaboration
but
yeah
we
learned
some
collaborating
work
about
the
validation
of
salary
using
various
animals
yeah
so
we
hope
we
can
show
some
interesting
results
following
results

using
animal
data
very
interesting
yeah
well
it speaks
a lot
to the
stage
that
our
field
is in
in
certain
ways
where
we've
seen a lot
of
graphics
that are
suggestive
this paper
and the
building on
the previous
2020 paper
made a
suggestive
possibility
much
much closer
to
a
analytically

demonstrated
translation
and then
took the
next step
incrementally
into the
in silico
agent
and so
it's
only natural
to then
explore
different
embodied
systems
as well
um
are there
any other
sections
that you
wanted to
like
look at
or highlight
or any
other
topics
about
the paper
or adjacencies
or active
inference
that you
think are
interesting

to go
into
yeah
so
okay
then
I would
like to
mention
about some
limitation
of this
paper
which is
not
directly
discussed
in the
papers
so
for example
well
we
focus
on
a
discrete
state
space
model
so
we
avoid
to
assume
some
substrate
that

encodes
the
covariance
of the
distribution
so
once you
assume
POMDP
then
it is
categorical
distribution
so
it is
different
from
assuming
Gaussian
distribution
characterized
by mean
and variance
so
the
neuronal
substrate
of
variance
is
still
unclear
and
we
now
try
to
figure

out
that
so
this
is
one
direction
of
limitation
and
another
limitation
is that
thanks to
a simple
simple
home
DP
structure
in

this
paper
we
don't
care
about
the
hierarchical
you know
optimization
but
generally
it is
crucial
to
update
parameters

through
hierarchical
optimization
through some
backpropagation
like
computation
although
it is
unclear
whether
backpropagation
itself
occurs
in the
real
brain
but
we
still
have
some
alternative
that
achieve
such
an
optimization
and
its
neuronal
substrate
is still
unclear
and
this
paper
does

address
that
direction
another
area
I'm
wondering
about
is like
where
in
neural
structures
is the
learning
reflected
where
is the
function
and
learning
reflected
well
sometimes
it has
to do
with
not
just
structural
tweaking
but
the
presence
or
absence
of
synapses

so
obviously
this
model
does not
expand
into
synaptogenesis
synaptic
pruning
let alone
neurogenesis
and
neuro
apoptosis
which
we mentioned
in the
previous
discussion
but
understanding
how
these
larger
scale
structural
changes
which
are
certainly
important
in
biological
systems
become
reflected
in

artificial
neural
networks
and
how
that
translates
all the
way
back
to
POMDP
and
whether
we
could
go
the
other
way
what
kinds
of
POMDP
structures
in
their
neural
network
realization
would
have
structural
modification
like
you
can
imagine

a
POMDP
that
does
structure
learning
but the
neural
network
doesn't
have
structural
change
or
there's
a
POMDP
that
doesn't
do
structural
learning
but it's
manifested
by a
neural
network
that
does
have
a
structural
change
element
so
structure
is
doing

something
very
different
in
these
two
different
categories
of
model
and
then
also
even
within
neural
firing
which
is
different
amongst
different
species
and so
on
there's
different
aspects
of what
that
firing
is
that
would
have
different
implications
for the

actual
biological
substrate
of
cognition
like
the
simple
connection
is
firing
rate
to
posterior
belief
average
firing
rate
no
change
in
posterior
reduce
the
firing
rate
if
the
posterior
should
be
going
down
increase
it
if
it
should

be
going
up
or
maybe
there
are
neurons
that
have
a
flipped
valence
but
the
same
type
of
relationship
but
there's
other
firing
patterns
like
spike
time
dependent
plasticity
synchronization
amongst
different
brain
regions
there's
a lot
of
things

that
don't
change
the
rate
overall
that
again
from the
biological
systems
we know
that those
phenomena
and
mechanism
are
important
for
different
cognitive
processes
so
there
will
be
many
many
years
of
a
fruitful
!
relationship
I'm
going to
bring
in this

picture
that
Alexandra
had taken
maybe we
need a
third
panel
in
figure
one
because
these
three
systems
moving
between
them
is going
to be
the
substance
of
the
field
for a
long
time
and
there's
maybe
other
edges
to
build
but
understanding
how

artificial
neural
networks
intermediates
between
the
empirical
measurements
and
manipulations
that we
can make
of real
neural
systems
and
the
interpretability
and
factorizability
of
POMDPs
it
might be
a bridge
too far
to go
from the
POMDP
to the
neuron
you could
always
use
this
technique
but it
would be

a purely
descriptive
statistic
type
approach
but it's
so interesting
that by
intermediating
through
a formal
connection
established
in
figure
one
I mean
in
equation
one
but also
shown here
then
we can
kind of
extend
the
chain
of
explanation
prediction
control
design
all the
way on
through
and that
just

unlocks
like an
incredible
amount of
neuroscience
that hasn't
been formalized
mathematically
and an
incredible
amount of
generative
models
that have
been specified
for different
learning
settings
sometimes
even
by analogy
to biological
settings
but
the metaphor
remains just
a metaphor
until it's
possible to
intermediate
with this
type of
neural network
development
yeah
that's a
crucial
point

it is
easy to
imagine
that
you know
real
data
or real
phenomena
can be
modeled
using
very
realistic
neural
model
or
glier
model
synaptic
model
right
it is
yeah
we believe
that it
is possible
and then
real
data
sorry
real
model
is not
not necessarily
tractable
not necessarily
useful

because it
is too
much
complicated
to
analyze
something
so we
use some
reduction
usually
mathematically
speaking
which
correspond to
topological
transformation
to make
the
model
simpler
and then
we need
to consider
the
translation
of that
simplified
neural
network
model
because
neural
network
model
itself
is
not

explainable
which
just
represents
some
dynamics
and its
functional
meaning
is
not
clear
but
thanks to
the
Bayesian
framework
we have
a very
nice
framework
to
lend
the
explainability
and
this
translation
this
correspondence
helped us
to
link
such
phenomena
based
equation
modeling

and
functional
based
equations
yeah
one
paper
from
2017
that
was
much
discussed
by
some
could
a
neuroscientist
understand
a
microprocessor
and
this
group
with
Jonas
and
Cording
they had
a
simulation
of
a
microprocessor
from
earlier
video game
console

I believe
and
then
using
the
analogy
of
like
the
transistors
and
their
connections
as
neural
firing
and
structural
connectivity
they were
able to
simulate
experimental
settings
input
and
action
and
then
make
measurements
from
like
every
neuron
including
doing
lesions

and loss
of
functions
and so
on
and it
turns out
that a
lot of
the
techniques
that
are
used
to
derive
scientific
explanation
from
analogous
data
collected
from
a
biological
system
those
techniques
which
ostensibly
should be
isolating
functional
explanations
in fact
did not
isolate
effective

explanations
you could
have a
deletion
over here
that induces
some
statistical
change
all the
way over
here
and that
may or
may not
be a
useful
clue
towards
the
function
of
even
sub
circuits
and so
I think
that
was a
wake-up
call
with
respect
to the
interpretability
of
simply
this

connection
between
the
biological
and the
neural
network
this
connection
alone
is of
limited
applicability
even
when the
neural
network
model
becomes so
complex
as to
recapitulate
the biological
phenomena
you're
never
under any
guarantee
that you're
going to
recover
interpretability
you may
have just
created an
atomic
level
simulation

of the
phenomena
but of
course a
map
that is the
same scale
as the
phenomena
isn't a
map
it's just
a copy
that has
no more
interpretability
and it's
almost like
what
what is
now
extended
again as
we kind
of just
summarize this
and think
about how
we move
forward
is that
connection
can now
be extended
into the
space of
interpretable
causal

models
and the
generalized
Bayesian
graphical
computational
frameworks
and all
the
heuristics
that we
can then
use like
variational
Bayes and
all these
other
methods
so it'd
be interesting
to look
back at
different
data sets
of in
vitro
and in
vivo
and in
silico
neural
activation
especially
if the
task
was of
this
constrained

set of
pomdps
and it
was already
amenable
because as
you brought
up other
settings
would require
a little
bit more
theory
development
before we
understand
what
pomdp
family
would be
applicable
cool
well
do you
have any
final
thoughts
or questions
well
do you
hop
I want
to
download
the
matlab
scripts
and

generate
the
figures
play
around
with a
few
of
these
parameters
!
like I
see that
you can
change
how far
the
entity
can
see
and
then
with
these
models
I'm
also
always
curious
about
the
computational
complexity
like if
you
extended
the
planning

horizon
from
four
to
five
or
you
dropped
it
down
to
three
what
is
the
runtime
consequences
and what
is the
performance
consequences
and where
might we
be able
to use
single
or swarms
of really
simple
agents
maybe even
making binary
decisions
and achieve
high
performance
and where
do we

really need
to
move
into
these
large
combinatoric
!
spaces
in order
to solve
problems
and the
kinds of
complex
planning
problems
that we
solve
whether
it's
planning
our
day
or
planning
our
week
are
those
more
like
true
deep
horizon
planning
problems
with

extensive
consideration
of
counterfactuals
and
calculation
of
alternatives
or
are
those
actually
composite
decisions
that are
made up
of
smaller
simpler
sub
decisions
that we
may or
may not
have
flexibility
to
restructure
so that
a decision
a complex
chess
maneuver
a sacrifice
in chess
or
another
game

it may be
possible to
model that
as
a deep
horizon
scan
or
a kind
of
intuitive
heuristic
for
inappropriate
skilled
entity
for
this
particular
two-layer
structure
there is a
clear limitation
about
the
horizon
of the
search
forward
search
prediction

doesn't
use forward
prediction
it's
used
post

diction
approach
so it's a
clear limitation
but still
it may have
some
have
performance
ability
to achieve
some level
of
performance
performance
that actually
I mean that
provides even
another
way to
look at
planning
so the
two ways
I was
describing
planning
as it's
often
described
in the
PMDP
literature
is again
is it a
true
deep
horizon

consideration

or

is it

just

short-term

heuristics

or

nested

models

that are

short

and I

think

that

this

paper

says

maybe

neither

maybe

it's

purely

the

fictive

causality

on the

past

that

leads

to the

emergence

of

sentience

and

maybe

even

teleological

planning

like
behavior
through
the
ongoing
reconsideration
of the
consequences
of past
action
but it's
neither
a short
nor a
long-term
planning
challenge
it's
actually
like a
memory
and
learning
challenge
and no
planning
occurs
right
but
indirect
to
planning
you know
planning
element
is
involving
C

matrix

planning

planning

planning

as a

phenomena

occurs

and

de-risking

through

time

occurs

but

there

but

it says

something

quite

interesting

and deep

that

that

phenotype

or function

could be

enacted

by a

system

that

explicitly

looks a

long way

ahead

explicitly

looks a

short way

ahead

or
moves
forward
and looks
backwards
only
right
which is
what they
sometimes
say about
the past
and the
future
so that
may be
a very
biologically
plausible
form
of
learning
and
it's
already
intimately
connected
with the
dynamics
and the
plasticity
in terms
of an
integrated
loss
function
so these
are all

excellent
directions
to keep
learning
on
right
and
I'm also
interested
in the
biological
implementation
of such
a short
term
or long
term
forward
prediction
and
planning
and I
hope
to find
some
nice
connectivity
to
such
an
implementation
of
Bayesian
model
and
implementation
in
real

brain
network
cool
and also
I'm
always
curious
about
the
invertebrate
brain
as a
ant
researcher
and so
many of
the
micro
architectures
as well
as the
regional
architectures
that people
discuss
are
mammalian
centric
which makes
sense
the mammalian
cortical
column
and the
relationship
with
the
dopaminergic

midbrain
and the
cortical
regions
and the
spinal
reflex
arc
those are
all
important
systems
of interest
yet
the
micro
and
meso
anatomy
of the
invertebrate
nervous
system
is
pretty
distinct
so
our
model
should
be able
to
describe
neural
and
cognitive
systems
of course

across
invertebrates
and
vertebrates
so I
look forward
to also
seeing
what those
models
of the
invertebrate
nervous
system
and
collective
behavior
where you
could have
some type
of
backwards
looking
risk
inference
of
the
swarm
who
knows
who
knows
well
we
really
appreciate
the time
that you

took for
these
discussions
I think
they
are
immensely
important
and we
wish you
the best
of luck
in these
continued
directions
yeah
okay
that's it
thank you
thank you
very much
see you
next time
bye
bye
thank you
Thank you.